

Simulation Evidence Weight Module (SEWM)

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Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: SIMULOS® — Simulation Governance Architecture

Parent Standard: Simulation Evidence & Reliance Boundary Standard (SERBS)

Module: SEWM — Simulation Evidence Weight Module

Operational Layer: Simulation Evidence Weight Layer

Category: Governance & Enforcement

Subcategory: Simulation Governance

Type: Core Module

Governed Space: Simulation Evidence Weight

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Canonical Definition

Simulation Evidence Weight Module (SEWM) is the second Core Module of the Simulation Evidence & Reliance Boundary Standard (SERBS) within SIMULOS® — Simulation Governance Architecture.

SEWM governs the simulation-specific evidential weight assigned to qualified Simulation Evidence by determining how strongly that evidence may legitimately support, challenge, constrain, compare, or inform a defined evidential proposition within its governed simulation context.

The Module evaluates the material factors that strengthen, weaken, condition, narrow, or qualify evidential significance, including reality representation, assumption dependence, model applicability, input representativeness, scenario relevance, execution basis, repetition structure, output meaning, inference distance, simulation-to-reality correlation, uncertainty, evidence dependence, and temporal relevance.

SEWM does not determine whether the complete evidence basis is sufficient for a Validation Determination.

Its task is narrower:

to determine how much simulation-specific evidential significance qualified Simulation Evidence may legitimately carry before reliance boundaries and

broader validation sufficiency are considered.

Core Question

How much evidential significance may qualified simulation evidence legitimately carry within its defined context?

Architectural Position

SEWM is the second module within SERBS.

It follows:

SEQM — Simulation Evidence Qualification Module

and precedes:

SRBM — Simulation Reliance Boundary Module.

SEQM establishes:

what legitimately qualifies as Simulation Evidence.

SEWM then establishes:

how much simulation-specific evidential weight that qualified evidence may carry.

SRBM subsequently determines:

how far reliance upon that evidence may legitimately extend.

The sequence is therefore:

Qualification → Weight → Reliance Boundary.

Internal Flow Position

Qualified Simulation Evidence → Evidential Proposition Alignment → Evidence Strength Factor Identification → Evidence Weakness Factor Identification → Dependence Assessment → Uncertainty Effect Assessment → Correlation Contribution Assessment → Contextual Weight Determination → Simulation Evidence Weight State → Weighted Simulation Evidence → SRBM — Simulation Reliance Boundary Module Operational Space

SEWM governs the space between:

qualified evidence

and

the evidential significance attributed to that evidence.

This operational space includes: evidence-to-proposition strength; representational contribution; assumption dependence; model applicability;

environmental applicability; input representativeness; scenario relevance; scenario coverage relevance; execution consistency; repetition value; evidence independence; common-mode weakness; output directness; inference distance; correlation contribution; uncertainty effects; contradictory evidence; evidence convergence; evidence divergence; temporal relevance; evidence degradation; contextual weight assignment.

Operational Scope

SEWM applies whenever qualified simulation evidence is expected to carry evidential significance beyond mere existence.

It may govern evidence derived from: individual simulation runs; repeated simulations; Monte Carlo studies; scenario portfolios; large simulation campaigns; comparative simulations; sensitivity studies; failure simulations; mission simulations; engineering simulations; safety simulations; scientific simulations; digital twins; infrastructure simulations; economic simulations; AI simulations; autonomous-system simulations; physical simulations; hybrid simulations; future simulation technologies.

The Module remains technology-neutral.

Its concern is not computational sophistication.

Its concern is:

how much evidential meaning the governed simulation basis can legitimately support. Evidence Qualification $\neq$ Evidence Weight

SEWM enforces the distinction:

Qualified Evidence $\neq$ Strong Evidence

SEQM may determine that an evidence object is:

Qualified

because it is sufficiently identified, traceable, contextualized, and interpretable.

SEWM may nevertheless determine that the evidence carries:

low evidential weight

because it depends heavily upon uncertain assumptions, weak reality representation, narrow scenarios, limited correlation, or other material constraints.

Conversely, conditionally qualified evidence may carry substantial evidential weight within a narrow context.

Qualification answers:

May this information function as evidence?

Weight answers:

How strongly may this evidence support the defined proposition?

Evidence Weight Is Proposition-Specific

Evidential weight does not exist in isolation.

The same simulation evidence may carry different weight for different propositions.

For example, a simulation may provide:

high evidential weight

for the proposition:

“Configuration A performs more efficiently than Configuration B under Scenario Set S.”

while providing:

low evidential weight

for the proposition:

“Configuration A will outperform Configuration B across unrestricted operational reality.”

SEWM therefore requires evidence weight to remain linked to the specific proposition being considered.

Representational Strength

SEWM evaluates how the governed reality representation affects evidential weight.

Relevant factors may include: whether materially relevant aspects of reality were represented; abstraction level; fidelity; resolution; excluded phenomena; proxy use; simplified interactions; representation limitations.

High representational sophistication does not automatically produce high evidential weight.

The relevant question is:

Was the reality necessary for this evidential proposition represented sufficiently?

Assumption Dependence

Simulation evidence may depend materially upon assumptions.

SEWM therefore evaluates: number of material assumptions; consequence of assumption failure; sensitivity to assumption variation; dependencies between assumptions; uncertainty surrounding assumptions; whether assumptions are empirically supported; whether results remain stable when assumptions change.

Evidence whose conclusion disappears under plausible assumption variation may

legitimately carry less weight.

Model Applicability

A simulation model may be technically valid within one range while poorly suited to another.

SEWM therefore considers whether: the model was used within its intended applicability; extrapolation occurred; relevant behaviors are represented; model simplifications affect the proposition; parameter ranges remain supported; model structure creates known systematic limitations.

Model sophistication $\neq$ model applicability.

Input Representativeness

Qualified evidence may still carry limited weight where its input basis poorly represents the evidential context.

SEWM may consider: source relevance; distribution representativeness; temporal representativeness; environmental representativeness; population representativeness; transformation effects; synthetic-data dependence; input uncertainty; missing input conditions.

The relevant question is:

Do the inputs support the evidential proposition being made?

Scenario Relevance

Evidence weight depends partly upon whether the simulated scenarios materially address the proposition.

SEWM evaluates: relevance of scenarios to the evidential question; coverage of expected conditions; representation of material variations; edge-case inclusion; stress-condition inclusion; uncertainty coverage; scenario omission; scenario overconcentration.

A very large simulation campaign may carry little weight for a specific proposition if the decisive scenario was never simulated.

Execution Basis

SEWM may consider whether evidential weight is strengthened or weakened by the execution basis.

Relevant factors may include: controlled execution; configuration consistency; execution reproducibility; runtime intervention; failed executions; execution anomalies; incomplete runs; execution drift; campaign consistency; reconstructability.

SEWM does not govern execution itself.

That remains under SES.

It uses governed execution information as an input into evidential weight

assessment.

Repetition Value Principle

Repeated simulation may increase evidential weight.

But repetition does not automatically do so.

SEWM distinguishes:

Repeated Evidence from Independent Evidential Support.

Repeated runs may increase weight where they legitimately test: stochastic variation; uncertainty; parameter variation; scenario diversity; sensitivity; reproducibility; robustness.

Repeated runs may add little weight where every execution reproduces the same material weakness.

Evidence Independence

SEWM requires evidence dependence to remain visible.

Evidence objects may share: the same model; the same assumptions; the same input source; the same scenario design; the same reality representation; the same execution infrastructure; the same correlation basis; the same transformation methodology.

Ten thousand evidence objects sharing one material weakness may not provide ten thousand independent confirmations.

Therefore:

Evidence Count $\neq$ Evidence Independence.

Common-Mode Weakness

A common-mode weakness is a material limitation shared across multiple evidence objects.

Examples include: one incorrect assumption; one omitted physical phenomenon; one unrepresentative dataset; one systematic sensor model error; one missing scenario class; one correlation deficiency; one shared transformation error.

SEWM requires common-mode weaknesses to reduce the apparent independence of aggregated evidence where materially relevant.

Output Directness

Evidence may carry different weight depending on how directly it arises from governed outputs.

SEWM distinguishes between:

Directly Observed Simulation Evidence

and

Derived or Inferred Simulation Evidence.

Direct evidence may include: observed state; threshold crossing; collision event; simulated failure; measured simulated parameter.

Derived evidence may require: aggregation; transformation; inference; statistical interpretation; causal interpretation; model-based extrapolation.

Derived evidence is not automatically weaker.

But the inferential pathway must be included when assessing weight.

Inference Distance

SEWM introduces Inference Distance as the methodological distance between:

what the simulation directly produced

and

what the evidence is being used to claim.

Short inference distance may occur where:

output → directly corresponding proposition.

Long inference distance may occur where:

output → transformation → aggregation → statistical inference → causal interpretation → real-world claim.

Greater inference distance may introduce additional dependence, assumptions, and uncertainty.

It must therefore remain visible in evidence-weight determination.

Correlation Contribution

Simulation-to-reality correlation established through SRCS may materially influence evidential weight.

SEWM may consider: strength of observed correlation; number of reality comparison points; diversity of comparison conditions; correlation uncertainty; known deviations; systematic discrepancies; temporal stability; correlation applicability to the evidential proposition.

However:

Correlation Strength ≠ Reality Equivalence.

Strong correlation may strengthen evidence.

It does not remove all remaining limitations.

Uncertainty Effect

SEWM requires material uncertainty to influence evidence weight.

Relevant uncertainty may arise from: inputs; assumptions; parameters; model structure; scenario selection; execution variability; outputs; inference; correlation; represented reality.

Uncertainty need not make evidence unusable.

It must make evidential weight more realistic.

Sensitivity Principle

Evidence weight should reflect the sensitivity of the evidential conclusion to material changes in the simulation basis.

Where small changes in: assumptions; inputs; parameters; scenarios; environmental conditions; model structure

produce major changes in the conclusion, evidence may require lower or more conditional weight.

Where the conclusion remains stable across materially relevant variation, evidence weight may be strengthened.

Evidence Convergence

Multiple evidence objects may increase evidential weight where they converge upon the same proposition through meaningfully different evidential pathways.

Convergence may be stronger where evidence differs across: models; assumptions; input sources; scenarios; methods; environmental configurations; execution structures; correlation sources.

SEWM therefore distinguishes:

Repeated Agreement

from

Independent Convergence.

Contradictory Evidence

SEWM must preserve material contradictory simulation evidence.

Evidence weight cannot legitimately be determined only from evidence supporting the preferred proposition.

Contradictory evidence may include: different model results; scenario-specific failures; alternative parameter outcomes; inconsistent repeated runs; operational disagreement; correlation failures; unexpected sensitivities.

Contradiction may: reduce evidential weight; narrow applicability; require conditional weighting; trigger further simulation; require separate evidence classes.

Evidence Weight States

SEWM may assign a governed evidence-weight state.

A minimum state model may include: 1. Indeterminate Weight

Evidence is qualified, but the simulation basis is insufficiently characterized to assign meaningful evidential weight. 2. Limited Weight

Evidence legitimately informs the proposition but is materially constrained by uncertainty, representation, assumptions, scenarios, correlation, dependence, or other weaknesses. 3. Moderate Weight

Evidence provides meaningful support within a defined context, while material limitations remain. 4. Substantial Weight

Evidence provides strong simulation-specific support for the defined proposition under the governed conditions. 5. Conditional Weight

Evidence may carry substantial or moderate weight only within explicitly defined conditions. 6. Degraded Weight

Previously assigned weight has materially weakened because of new information, reality change, model change, evidence contradiction, correlation deterioration, or another material change.

The states are not universal scientific truth categories.

They are governance states requiring defined organizational criteria. Weight State ≠ Validation Status

An evidence object with Substantial Weight is not automatically sufficient for validation.

Likewise, multiple substantial-weight evidence objects do not automatically establish validation sufficiency.

SEWM governs:

simulation-specific evidential significance.

VALIDOS® governs:

the sufficiency of the total validation basis.

Quantitative and Qualitative Weighting

SEWM may support: qualitative weighting; ordinal weighting; numerical weighting; probabilistic weighting; multi-factor weighting; evidence matrices; structured expert assessment; hybrid approaches.

SIMULOS® does not require one universal scoring method.

The weighting method must remain: explicit; interpretable; proposition-linked; context-aware; reproducible where appropriate; transparent about uncertainty; bounded by its intended use.

A numerical score must not create false precision.

Weight Calibration

Organizations may establish evidence-weight criteria appropriate to their context.

Calibration should consider: consequence of reliance; simulation type; evidential purpose; available reality data; expected uncertainty; model maturity; scenario criticality; safety relevance; operational significance.

SEWM governs the simulation-specific structure of weighting.

It does not establish organization-wide risk appetite or accountability.

Temporal Weight

Evidence weight may change over time.

Weight may increase where: new real-world correlation becomes available; additional independent evidence converges; model limitations are reduced; relevant scenario coverage expands.

Weight may decrease where: reality changes; the Simulation Object changes; assumptions become obsolete; contradictory evidence emerges; correlation deteriorates; new failure modes are discovered.

Evidence weight must therefore be reviewable rather than permanently fixed.

SEWM Boundary with SEQM

SEQM determines whether simulation-derived information legitimately qualifies as evidence.

SEWM determines how much evidential weight the qualified evidence may carry.

Therefore:

Qualification $\neq$ Weight.

SEWM must not retroactively redefine evidence qualification except where weight assessment reveals missing information that requires SEQM reassessment.

SEWM Boundary with SRBM

SEWM determines:

how strong the evidence is within its governed context.

SRBM determines:

how far reliance upon that evidence may extend.

A strong evidence object may still have a narrow reliance boundary.

Therefore:

High Weight ≠ Broad Reliance.

SEWM Boundary with VALIDOS®

SEWM does not determine: total evidence sufficiency; validation criteria satisfaction; Validation Determination; operational reliance authorization.

VALIDOS® remains responsible for broader validation governance.

The boundary is:

SEWM: How much simulation-specific evidential significance may this evidence carry?

VALIDOS®: Is the total evidence basis sufficient for the relevant validation claim and intended reliance?

SEWM Boundary with SRCS

SRCS determines the governed correlation relationship between simulation and reality.

SEWM may use correlation as one evidential-strength factor.

SEWM does not establish correlation independently.

SEWM Boundary with SSS

SSS governs scenario design and scenario coverage.

SEWM may assess how scenario relevance and coverage affect evidential weight.

It does not redesign or retroactively redefine the scenario space.

SEWM Boundary with SAS

SAS governs simulation assumptions.

SEWM evaluates how dependence upon those governed assumptions affects evidential weight.

It does not determine the universal validity of the assumptions.

SEWM Boundary with SIDS

SIDS governs simulation inputs and data at the simulation-input boundary.

SEWM evaluates how input representativeness and uncertainty affect evidential significance.

It does not become a general data-quality framework.

SEWM Boundary with INTEGROS®

SEWM does not determine general evidence integrity.

It assumes governed integrity information may be relevant to evidence-weight assessment where supplied by INTEGROS®.

Integrity and evidential weight remain distinct governance questions.

SEWM Boundary with ADIT®

ADIT® may determine whether evidential records and execution evidence are sufficiently observable, preserved, verifiable, or auditable.

SEWM determines the simulation-specific evidential weight those governed records may support.

Auditability ≠ Evidential Weight.

Minimum Implementation Framework

A minimum SEWM implementation should establish: 1. Qualified Evidence Input

Receive a qualified Simulation Evidence object from SEQM. 2. Evidential Proposition

Confirm the specific proposition or question against which weight is being assessed. 3. Strength Factor Identification

Identify material factors strengthening evidential meaning. 4. Weakness Factor Identification

Identify material factors weakening or constraining evidential meaning. 5. Dependence Assessment

Identify dependencies across evidence objects and common-mode weaknesses. 6. Uncertainty Assessment

Determine how material uncertainty affects evidential significance. 7. Correlation Contribution

Determine how governed simulation-to-reality correlation affects evidence weight where relevant. 8. Contradictory Evidence Review

Identify materially conflicting or divergent evidence. 9. Weight Determination

Assign a governed simulation-specific evidence-weight state or value. 10. Weight Basis Record

Document the basis, conditions, uncertainty, dependencies, and factors supporting the determination.

Governance Outputs

SEWM may produce: Simulation Evidence Weight Assessment Evidence Strength Factor Record Evidence Weakness Factor Record Assumption Dependence Record Scenario Relevance Assessment Evidence Independence Assessment Common-Mode Weakness Record Inference Distance Record Correlation Contribution Record Evidence Uncertainty Effect Record Contradictory Evidence Record Evidence Convergence Record Simulation Evidence Weight State Simulation Evidence Weight Basis Weighted Simulation Evidence Object

These outputs provide the governed basis for:

SRBM — Simulation Reliance Boundary Module.

Enterprise Application

SEWM enables organizations to solve a problem that appears repeatedly in large-scale simulation programs:

a simulation may generate enormous quantities of qualified evidence without all of that evidence carrying equal significance.

An organization may use SEWM to determine: whether repeated simulations provide genuinely new evidence; whether millions of runs share one common model weakness; whether evidence depends critically upon one assumption; whether scenario coverage actually addresses the proposition being supported; whether correlation with reality meaningfully strengthens evidence; whether an aggregate success rate conceals important failures; whether multiple models independently converge; whether inference from output to real-world claim is short or extensive; whether uncertainty materially weakens the evidential conclusion; whether previously strong evidence has degraded after reality change; whether evidence should carry limited, moderate, substantial, conditional, indeterminate, or degraded weight.

SEWM therefore prevents organizations from equating:

more simulations

with

more proof.

A billion simulations using the same incomplete representation may provide less evidential strength than a smaller body of independently convergent simulations supported by strong reality correlation and materially different evidential pathways.

The Module gives organizations a method for distinguishing evidence volume from evidence strength.

Foundational Principles

Qualified evidence is not automatically strong evidence.

Evidence weight is proposition-specific.

Evidence volume is not evidential strength.

Repeated evidence is not automatically independent evidence.

Common assumptions create common evidential dependencies.

Simulation sophistication does not determine evidence weight.

Numerical precision does not create evidential certainty.

Scenario relevance matters more than scenario count.

Correlation may strengthen evidence without creating reality equivalence.

Independent convergence may carry greater evidential significance than repeated agreement generated through one common pathway.

High evidential weight does not automatically create broad reliance.

Simulation evidence should carry only the weight that its governed representation, assumptions, model, inputs, scenarios, execution basis, outputs, correlation, dependencies, and uncertainty can legitimately support.

Canonical Closing Statement

Simulation Evidence Weight Module establishes the evidential significance layer within the Simulation Evidence & Reliance Boundary Standard by determining how strongly qualified Simulation Evidence may legitimately support, challenge, constrain, compare, or inform a defined proposition within its governed simulation context. Through governance of representational strength, assumption dependence, model applicability, input representativeness, scenario relevance, execution basis, repetition value, evidence independence, common-mode weakness, output directness, inference distance, correlation contribution, uncertainty, sensitivity, convergence, contradiction, temporal relevance, and evidence-weight states, SEWM prevents simulation evidence from acquiring strength merely through scale, repetition, complexity, numerical precision, or apparent sophistication. By separating evidence qualification from evidential weight, and evidential weight from reliance boundaries and validation sufficiency, SEWM preserves the methodological boundaries of SERBS and VALIDOS® while creating the governed basis required for downstream reliance control. Simulation evidence should carry neither more nor less evidential significance than the governed simulation conditions can legitimately support.

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